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PRACTICE MANAGEMENT

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HOW-TO GUIDE

Set up a Crystal Meet room

Put a Crystal Meet device in a conference room and join Zoom and Google Meet calls from the room's screen, step by step.

This guide takes one conference room from an empty Raspberry Pi to a room that shows up in the Crystal Meet dashboard and joins calls from its own web page. Plan about 45 minutes per room, most of it waiting for downloads. No programming experience is needed: if you can follow a recipe and type the commands printed here, you can do this. Repeat it for each room. If the room's TV is an LG or Samsung, the optional section The TV, after step 4, sets up its controls on the table screen.

STEP 1
Prepare the device
Install Raspberry Pi OS, sign in, connect the TV and camera.

STEP 2
Create the room
Add the room on the dashboard's **Provisioning** page and copy its token.

STEP 3
Install Crystal Meet
One command with the room's config file.

STEP 4
First run and screens
Join a test call, pick the TV's screensaver, then point the table screen and the door sign at the device.

BEFORE YOU BEGIN

Hardware per room. Raspberry Pi 4 (4 GB) or Pi 5, the official power supply, a 32 GB microSD card, an HDMI cable to the room's TV, and a USB camera and speakerphone (for example a Logitech Brio 505 and a Jabra Speak2 55).

Two screens on the network. A touch tablet or PoE touch panel on the table to join and control meetings, and a small screen by the door to show whether the room is free. Neither connects to the Pi: each only needs a browser, power, and the office network. Give the room device a fixed address on your router (a DHCP reservation) so the screens can always find it.

The dashboard. The Crystal Meet dashboard must already be running on the office network; the guide "Set up the Crystal Meet dashboard" installs it on its own Raspberry Pi with a fixed address, for example `http://10.0.0.50`. Rooms talk to it on port `3001`.

Accounts. A dashboard sign-in with the admin role, and the Zoom or Google Meet links you want to test with.

1 Prepare the device

- **Flash the card.** On your computer, open **Raspberry Pi Imager**, choose your Pi model, then **Raspberry Pi OS (64-bit)** with desktop, and your microSD card.
- **Pre-fill the settings.** When Imager offers customisation, set the hostname to something like `crystal-meet-room-1`, the username to `pi` with a password you keep, your Wi-Fi network and your time zone, and turn on SSH. Write the card.
- **Connect and boot.** Put the card in the Pi, plug in the TV over HDMI, the camera and speakerphone over USB, then power. Wait for the desktop to appear on the TV.
- **Turn on automatic sign-in.** Open **Preferences › Raspberry Pi Configuration › System** and set **Auto Login** to on, so the room comes back by itself after a power cut.

You should now see the Raspberry Pi desktop on the room's TV, and the Pi should be on the office network.

2 Create the room in the dashboard

- **Open the dashboard** at its address, for example `http://10.0.0.50` or `http://crystal-meet.local`, and sign in.
- **Go to Provisioning** and fill in the **Room Name** exactly as people call the room (for example `Room 1`) and, if you like, its **Location**. Press **Generate Token**.
- **Copy the token** shown under **Enrollment Token**. It is a long string of letters and numbers; you will paste it into the device's config in the next step.

You should now see the token on the page and, on **Devices**, the room listed as provisioning until its device connects.

> **GOOD TO KNOW**
>
> The token links one device to this room and works once. If a device is ever reinstalled, create a new token for it. Nothing else about the room needs to change.

3 Install Crystal Meet on the device

- **Open a terminal** on the Pi (the black icon in the top bar), or connect over SSH from your computer.
- **Get the software.** Type `sudo apt update && sudo apt install -y git`, then `git clone https://github.com/ben-abeo/croom.to.git`, then `cd croom.to`.
- **Make the room's config.** Type `cp deploy/rooms/room-1.yaml ~/room.yaml` (use `room-2.yaml` or `room-3.yaml` for the other rooms), then `nano ~/room.yaml`.
- **Fill in three values.** The room's **name** and, if people call the place something, its **location** (delete that line to show the name alone); the dashboard **url** as `http://<dashboard address>:3001`; and the **enrollment_token** you copied. Save with **Ctrl+O**, Enter, then **Ctrl+X**.
- **Run the installer.** Type `sudo bash installer/install.sh --config ~/room.yaml`. It installs the browser and the room software; expect 10 to 20 minutes.
- **Start it.** When the installer prints **Crystal Meet installation complete**, type `sudo systemctl start croom`. The service also starts on every boot from now on. If it ever stops, the **Start Crystal Meet** icon on the device's desktop (over VNC) starts it again.

  You should now see the room's screensaver fill the TV: the room's name, its status and the time. The room's TV needs no keyboard or mouse, and nobody has to touch it.

4 First run and the screens

- **Open the room page.** On your laptop or phone on the office network, go to `http://crystal-meet-room-1.local:8080`, or use the device's IP address with `:8080`.
- **Check the dashboard.** On **Devices**, the room should show **online** within a minute.
- **Join a test call.** Paste a Zoom link, or a Zoom meeting ID, into **Join with a link** and press **Join**. A meeting ID, or a Zoom link without its passcode, makes a **Passcode** field appear under the box: type the meeting's passcode there before you press **Join** (on the table screen, tapping the field opens the on-screen keyboard on its digits). The TV shows the meeting joining; the page switches to the meeting screen, which offers **Mute**, **Turn camera off** and **Leave** and, depending on the platform, more (see **The meeting screen** below).
- **Leave.** Press **Leave**, then **Tap again to leave**. The TV returns to the screensaver.
- **Pick the TV's idle look.** Tap **Options**, then **Screensaver** on the room page, and choose **Information** (name, status, next booking and the time), **Quiet** (name and status only), **Brand** (the logo) or **Bounce** (the logo drifting around the screen). The TV changes at once and the choice is remembered.
- **Connect the calendar.** To show real bookings and let people join them, follow the guide **Connect Crystal Meet rooms to Google Calendar** next. Until then the room page works with pasted links only.
- **Connect Zoom.** To join Zoom meetings, including ones hosted by other companies, follow the guide **Connect Crystal Meet rooms to Zoom**. Without it, Zoom turns the room away as an automated guest.

- **Set up the table screen.** On a Raspberry Pi with the touch display, clone the repository as in step 3 and type `sudo bash installer/install-kiosk.sh --url http://<device address>:8080/`. The room page opens full screen at every login with its own on-screen keyboard, and a **Room controls** icon on the desktop reopens it if it is ever closed; run the command again to change the address. After updating Crystal Meet, pull the repository on the table Pi too, run the install-kiosk command again and reboot it so the new scale takes effect. On a tablet, open its kiosk browser (Fully Kiosk Browser on Android, Guided Access with Safari on an iPad), set the start page to `http://<device address>:8080/`, turn on full screen and keep the screen awake.
- **Type on the table screen.** Tap the link box (or the passcode field) and the on-screen keyboard comes up from the bottom of the screen. It has three layers, like an iPad's: letters first, then **123** for numbers and the symbols a link or a passcode needs, then **#+=** for the rest; **ABC** goes back to the letters. Tap the shift key (the up arrow) once for one capital, or twice quickly to lock capitals on, as caps lock does, and a dot shows under its arrow; tap it again to let go. Every key answers the moment a finger lands, so both thumbs can type at once, and holding the delete key keeps deleting. Each key clicks softly, and a letter or symbol shows a larger copy of itself above the key while you press it. **Join** on the keyboard joins the meeting you typed; the key with the down arrow puts the keyboard away, and so does a tap anywhere else on the page. To turn the click off, tap **Options**, then **Key click off** under **Keyboard**; **Key click on** brings it back. The room remembers the choice, for every screen that opens its page.
- **Set up the door sign.** On the door screen, set the start page to `http://<device address>:8080/sign`. It shows the room's name, its status and today's next bookings over a photograph of Austin, with a bar down the left edge and a label under the room's name that are green when the room is free, amber when a meeting is about to start, and red while the room is in use or booked. The section "The door sign" after this step says what else it shows and how to choose its background and turn on football scores. On an SMT1019 wall panel, install the Crystal Meet Door Sign app instead, as the guide "Set up a Crystal Meet door sign panel" describes: it shows the same page and lights the bar round the panel in the status colour.
- **Set up the camera and sound.** Tap **Options**, then **Camera framing** (the TV shows the camera between meetings; a screen left alone returns to the idle screen after ten minutes, and **Back** returns at once), press **Find the stops**, nudge with the arrows and zoom, then **Save as Home**. Frame the table, press **Save**, then a preset; repeat for the whiteboard. Tap a preset to return to it, even in a call. Moves take seconds (**Moving…**); the camera goes Home at every service start. During a call the page shows the meeting screen on its own, as a control deck: on the left the meeting's card, with the time in the call, who is in it, the chat that waits and **Leave** at its foot; on the right **Mute**, **Turn camera off**, **Raise hand** and **Reactions** as tiles, then **Captions**, the view, **Hide self view**, **Participants**, **Chat**, **Host** (for a host) (the section after this step shows them all), and at the foot the volume, **Camera** for a preset or a nudge, and **Upcoming meetings** to see what is booked next, with **Touchpad** and **TV** in the top bar. On that screen, under **Sound**, set the level and try **Mute speaker**; `audio.output_device: HDMI` in the room config (then restart) sends the sound to the TV. **No speaker found** or **No controllable camera found**: the device sees no speaker, or no camera that pans or zooms (the MeetUp does).

  You should now see the room page say the room is free again, the door sign's bar and label green, and the dashboard still showing the device online.

TIP

Zoom links that include their passcode (the part after `?pwd=`) join straight through. Google Meet needs the room signed in once as its own Workspace user; the guide "Let Crystal Meet rooms join Google Meet" covers it.

ON THE TABLE

The meeting screen

While a call lasts the page shows the meeting screen on its own, and draws only what the call's platform offers the room right now. This is the screen at its fullest, in a Zoom meeting where the room is the host. A Google Meet call offers Mute, Turn camera off, Raise hand, Captions and Leave, and no more (besides **TV**, which every call has). Every button has an icon before its label (a dialog's card, below, is the exception); where a button shows a state, the icon is the state and the words are what a tap does.

The first row

Mute or **Unmute**

The microphone as the meeting has it: a host muting the room changes the button within a few seconds, and a tap asks for the state the button names, so a Mute tapped just after the host muted the room leaves it muted.

Turn camera off or **Turn camera on**

The same for the camera: a Turn camera off tapped after the camera went off at the TV leaves it off.

Raise hand or **Lower hand**

Zoom keeps the hand in its own Reactions list; the room presses it there.

Reactions

Opens a row for ten seconds: Thumbs up, Clap, Heart, Laugh, Wow, Celebrate, and Cancel.

Leave

Asks for a second tap: Tap again to leave.

The second row

Captions on or **Captions off**

Zoom's captions on the TV.

Gallery view or **Speaker view**

The button names the view a tap switches to.

Hide self view or **Show self view**

The room's own picture on the TV.

Participants (7)

Opens the list of people; the count leaves out the waiting room.

Chat (2)
Opens the chat; the number is the messages not yet read.

Host
Only while the room is host or co-host.

TV
The remote and touchpad of the room's TV, always the last button of the row: the section The TV, below, sets it up, and a room with no TV set up finds how to do it there.

The screens behind the second row

Participants
Everyone in the meeting, marked Host, Hand raised, Muted or Video off; the room says "(this room)". A host or co-host also gets Mute (only for someone not already muted) and Remove on each other person (Tap again to remove), and, under Waiting room, Admit for each person waiting.

Chat
The last 20 messages other people sent, newest at the bottom, with sender and time; No messages yet when there are none. It is read-only: there is no reply box.

Host controls
Mute all, Ask all to unmute, Start recording or Stop recording, Lock meeting or Unlock meeting, and, for the host only, End meeting for everyone (Tap again to end).

Each of the three has **Back**, and each goes back to the meeting screen on its own after two idle minutes; Camera and Upcoming meetings stay where you leave them. A tap anywhere on the screen starts the two minutes again.

What the meeting says to the room

- **The notice line.** A line under the meeting's title says what the meeting has disclosed of itself (recorded, transcribed, AI Companion) and stays for the rest of the call. It is two lines at most, and a longer one is cut short with an ellipsis.
- **The dialog card.** A Zoom dialog that waits for a tap is a card above the controls, with Zoom's own words and buttons in Zoom's order, the last one highlighted (Zoom puts the cancelling button first). The card's buttons carry no icons, because the words are Zoom's; its buttons ignore a tap for half a second after it appears, so a tap meant for the controls under it answers nothing by mistake. The Zoom guide says which dialogs wait and which are pressed for the room.
- **Second taps.** Leave, Remove and End meeting for everyone ask for a second tap: Tap again to leave, Tap again to remove, Tap again to end. Each must come within five seconds, and a double tap on the glass does not count as the second.

Icons are from Lucide (lucide.dev), ISC licence: the symbols above are the ones the table draws.

OPTIONAL, FOR AN LG OR SAMSUNG TV

The TV

The table screen can control an LG webOS TV or a Samsung Tizen TV, including the S90HD OLED, during and outside meetings. The TV Pi and the TV must be on the same local network. Set the TV up once from the table. Every room also has **Room Pi mouse** and **Full-screen touchpad** for controlling the Pi behind the TV, even when no TV is paired.

Set it up

- **On the TV, once.** For an LG TV, with the TV's own remote, open **Settings › General › Devices** and turn on **Mobile TV On** (some TVs call it Turn on via Wi-Fi) and **LG Connect Apps**. On recent webOS versions (the rooms' LG C6 OLEDs run webOS 26) they are usually under **TV Management** or **External Devices**; menus move between versions, so if yours differs, look through the TV's settings for those two names. Turn on **Quick Start+** as well if the TV has it: the TV then keeps its network connection while it is off, and the device can tell it to turn on over that connection, so Mobile TV On may stay off. Without Quick Start+ a TV that is fully off can only be woken by Wake-on-LAN, a wake-up message sent over the network, which needs Mobile TV On. The device always sends that message first, so the TV's MAC address must be saved either way.
- **For a Samsung TV.** Turn on **Power On with Mobile** in the TV's network settings so the Pi can wake it. Under **Device Connection Manager**, set **Access Notification** to **First Time Only** if the TV keeps asking to allow Crystal Meet. Menu names vary with firmware. Pair while the TV is on, with its remote nearby.
- **Find and select your TV.** On the table screen tap **Options**, then **TV**, then **Find TVs**. Select your LG or Samsung from the list. Crystal Meet saves its brand, IP address and the MAC address it could discover, then starts pairing immediately. Accept the request on the TV with its own remote within a minute. No config-file edit or service restart is needed. Reserve the TV's address on your router so it stays the same.
- **Check the connection settings.** The room assumes the Pi uses HDMI 1. Change that on the table if its cable is in another socket. If the MAC address could not be discovered, enter the address shown in the TV's network settings on the table; use the address for its active Ethernet or Wi-Fi connection. You can connect while it is missing, but turning an off TV on needs that address.

  The selection is saved in `tv-settings.json` in the room Pi's data folder, normally `/var/lib/croom`, and restored after a restart. These saved connection settings override the YAML defaults. Find TVs also lets you select a different TV later.

Optional manual defaults

Existing YAML configurations still work. For a room configured in advance, the optional block below sets LG defaults; add `brand: samsung` for Samsung. Replace the example addresses with the TV's own. Selecting a TV from the table takes precedence over these connection defaults.

```
tv:
  enabled: true
  host: "10.0.0.41"
  mac: "AA:BB:CC:DD:EE:FF"
  pi_input: "HDMI_1"
  keep_on: true
```

- **What the lines mean.** host is the TV's address. mac is its MAC address, which every Turn on needs, with Quick Start+ too; keep the quotes, because YAML reads a MAC address of digits only, such as `001122334455`, as a number. pi_input is the TV input the room device's HDMI cable is plugged into, `HDMI_1` for the first socket and `HDMI_2` for the second (the check below lists the TV's inputs). keep_on turns the TV on when the service starts, before every join and about 5 seconds after anything but the table turns it off (its own remote, for example); before every join it also switches the TV to pi_input, even from live TV or an app, and leaves a TV alone whose state it cannot read. A join waits up to 15 seconds for the TV to wake, with the page showing Joining, and then goes ahead without it. An Off from the table holds until someone presses On or the next join begins, and restarting the service forgets it.
- **Pair from the table.** Selecting a discovered TV starts this automatically. For an existing manual configuration, or after a pairing was declined, tap Pair, then accept the request that appears on the TV, with the TV's own remote, within a minute. The Pi stores each selected TV's pairing key privately in its data folder, normally `/var/lib/croom`. Old LG configurations keep using `tv-client-key` until a TV is selected on the table.
- **Check it from the device.** Type `/opt/croom/venv/bin/croom --check-tv -c /etc/croom/config.yaml` as the user the service runs as, the one who ran the installer (from another account, put `sudo -u pi` in front, with that user's name), and never as root: the TV's key file is private to that user (mode 600), and a check run as root would leave a renewed key where the service cannot read it. It reads:

```
TV: 10.0.0.41, MAC AA:BB:CC:DD:EE:FF
The TV is paired with this device (its key is in /var/lib/croom/tv-client-key).
The TV answers: LG C6, on.
Inputs: HDMI 1 (HDMI_1), HDMI 2 (HDMI_2).
The TV shows HDMI 1 (HDMI_1), the room Pi's input (tv.pi_input).
Ready: the TV answers.
```

- **What else it says.** The check only looks: it does not wake or pair the TV, and leaves nothing connected. A line beginning Warning (a MAC address that is missing or is not one, a pi_input the TV does not have, a TV showing another input) means the TV answers but something will not work as set, and the last line then reads "The TV answers, with the warnings above." When the TV does not answer, the check says why: it is not paired for this user, it does not answer, or it no longer accepts the device. If the TV shows its pairing prompt during the check and someone accepts it, the TV hands out a new key; the check keeps it and says "The TV issued a new key (its pairing prompt was accepted during the check), which is kept." It exits with 0 when the TV answered and 1 when it did not. With no saved or configured TV it lists the LG and Samsung TVs it finds on the network instead, as Find TVs does.

You should now see the check end with "Ready: the TV answers." and the table's remote open on the TV's name.

Using it

On Samsung, **Input** opens the TV's source menu; use the arrows and OK to choose the Pi's HDMI input. This integration does not report Samsung's current input, exact volume or mute state, so it does not automatically switch HDMI on a meeting join or show an invented volume. **Mute** toggles the TV's sound. The Samsung TV cursor is unavailable; **Room Pi mouse** and **Full-screen touchpad** still control the Pi. The following direct input and TV-pointer instructions apply to LG.

- **Open the remote.** Tap **Options**, then **TV**. During a meeting tap **TV** in the top bar. Beside it, **Touchpad** opens the full-screen touchpad on the room Pi's mouse in one tap, for a dialog on the TV in a call; **Done** returns to the meeting. **Back** returns to the screen you came from, and the remote returns by itself after ten minutes with no touch, or two minutes during a meeting.
- **What is on it.** The top line names the TV and says what it shows, for example **Boardroom TV · On · HDMI 1 (Crystal Meet)**. The name is the one the TV gives over its control connection, often its model name such as LG C6, so it can differ from the name Find TVs listed for the same TV. Under it are **Power**, **Input**, **Home** and **Menu**; the volume with **Mute**; the channel keys; a round d-pad with **OK** and, below it, **Back**, **Exit**, **Info** and **Guide**; the number keys with **Go**; the playback keys and the four colour keys.
- **Power, Input and the second taps.** **Power** asks for a second tap to turn the TV off, as Leave does. When the TV is off, the top line shows **Turn on** in place of its state, and turning it on takes up to about 23 seconds (the line says Turning on…); the other keys wait for it. **Input** opens a row of the TV's inputs, with the room device's marked **Crystal Meet**; a pick switches the TV at once. A key the TV refuses flashes amber once and the reason appears under the top line. **Forget TV**, at the right end of the top line, asks for a second tap too.
- **The touchpad modes.** Three buttons sit under the keys. **Pointer** turns the d-pad into a trackpad for the TV's own cursor: slide one finger to move it, tap to click, slide two fingers to scroll, and press and hold, then move, to drag. Tap **Pointer** again to bring the d-pad back. **Room Pi mouse** does the same for the Pi behind the TV: the Pi's desktop takes its virtual mouse like a USB mouse, so its pointer should appear on the TV (the room's first use confirms it), and you can click a button on whatever the Pi is showing, such as a dialog in a meeting. The TV shows the pointer only while the touchpad is driving the Pi's mouse, and for five seconds after: on the screensaver and in a meeting it is hidden. A two-finger tap there is a right-click.
- **The pad's border and its link.** A blue border on the pad (on the full-screen touchpad, a thin blue line round the screen's edge) means a hold is armed: after a short rest the next move drags, a touch that moves sooner is a move for good, and after a rest a nudge under 8 pixels does nothing. If the link to the device drops, the pad (the full-screen strip too) says **Reconnecting…** and the page tries again after 1 second, then 2, then every 5 seconds; a finger moves nothing until it is back.
- **The full-screen touchpad.** **Full-screen touchpad** gives the whole table screen to the pointer, with the same gestures. A strip names what it moves, with a switch between **TV** and **Room Pi** and a **Done** button; nothing else is on the screen to be tapped by mistake. It returns to the remote when you tap **Done**, or by itself after ten minutes with no touch. How the gestures feel, how far the cursor runs for a finger's travel and how long a tap may last, is tuned on the table's own screen.

FOR THE OPERATOR

A pairing waits 60 seconds for the person at the TV and gives the TV 5 seconds more to finish; Cancel, or Forget TV, ends it at once. While a TV is waking, which a join gives up to 15 seconds, the device sends the wake-up message again every 3 seconds. Restarting the service ends a pairing, a Turn on and an open touchpad within seconds, because it no longer waits for the TV. A TV that is not answering is reported in words under the remote's top line, for example "The TV is not answering at 10.0.0.41". Unpair with Forget TV, not by deleting the key file: a pointer that is already connected keeps working until its link drops. The touchpad's events travel over a WebSocket at `/ws/pointer` on port 8080, which like the rest of the room page takes no sign-in.

FOR EVERY ROOM DEVICE

Updating a room device

Update a device that already runs Crystal Meet in the order below. Every room device needs these steps, with an LG TV or not, because the library behind the touchpad's mouse is a hard dependency and is compiled on the device. The first and third steps are needed once and do no harm when repeated.

- **Install the build tools.** Type `sudo apt-get install -y python3-dev gcc libc6-dev`. Without them the update stops with an error while it builds the mouse library: the compiler's package only recommends libc6-dev, which has the C headers the build reads.
- **Update the software.** Type `sudo /opt/croom/venv/bin/pip install -q --force-reinstall --no-deps git+https://github.com/ben-abeo/croom.to.git`, then `sudo /opt/croom/venv/bin/pip install -q git+https://github.com/ben-abeo/croom.to.git`. The first replaces the software even though its version number has not changed (a plain upgrade would keep the old one); the second adds any library the new version needs.
- **Set the virtual mouse up.** Type `cd ~/croom.to && git pull`, then `sudo bash -c 'source installer/install.sh && install_virtual_mouse'`. That is the installer's virtual-mouse step on its own: a udev rule hands /dev/uinput to the input group, the uinput module is set to load at every boot, and your user joins the input group. (Running the whole installer again includes it.)
- **Restart.** Type `sudo systemctl restart croom`: a new group only reaches a service that starts afresh.

THE DOOR SCREEN

The door sign

The door sign is the page at `http://<device address>:8080/sign`, which step 4 pointed the door screen at (on an SMT1019 wall panel, the Crystal Meet Door Sign app shows it and lights the panel's bar green, yellow or red with it). Everything about it is chosen on the table screen, except **This week**, which is a button on the door sign itself: the door screen needs to be a touch screen to use it.

- **What it shows.** A photograph of Austin fills the screen under a navy overlay. Over it are the practice's name beside the logo, the time and the date, the room's name and where it is, and today's next three bookings, each with its time, its title and, for a video meeting, Zoom or Google Meet. The room's status is shown three ways at once, so it can be read from the hallway: a coloured bar down the left edge, a coloured label under the room's name that says it in words, and a ring round the booking in progress. Green is free (**Free until 1:00 PM**). Amber is the ten minutes before a booking (**Customer call starts in 8 minutes**). Red is in use or booked (**In use until 2:00 PM** while the room is in a meeting, **Booked until 2:00 PM** while a booking is under way). On a screen 560 pixels high or less the label before a booking leaves out the booking's title and says **Starts in 8 minutes**: Today shows the title beside it. The photograph itself is never tinted. A sign that cannot reach the device goes grey and says **Sign not connected**.
- **The practice's name.** The name beside the logo is **Crystal Practice Management**. To show another, type `sudo nano /etc/croom/config.yaml` and look under `room:`. If there is a `company:` line, change the name inside its quotes; if not, add a line under `name:` that begins with two spaces, `company: "Lone Star Eye Care"`, so that it lines up with the lines around it. Save, and type `sudo systemctl restart croom`. A `company: ""` line leaves the logo on its own.
- **Choose the background.** On the table screen tap **Options**, then **Door sign**. Under **Door sign background** are nine buttons: **Austin skyline** (the default), **Capitol**, **UT Tower**, **Pennybacker Bridge**, **Hamilton Pool**, **Texas star**, **Hill Country**, **DKR stadium** and **Plain**, which leaves the navy on its own. The one in use is the blue button. The door sign changes within a few seconds. The choice is kept on the device, so it survives a restart, and every door sign of the room follows it. If the sign is plain navy where a photograph should be, check first that **Plain** is not the one in use. A photograph is loaded when the sign opens, so a first load, or a reload, shows navy for a moment (about a third of a second, a second or more on a slow connection). A photograph that has not loaded after 20 seconds is given up on, and the sign stays navy, with a soft glow in the status colour in a corner, until the page is reloaded or the sign has lost the device and found it again.

- **See the week.** Tap **This week** beside today's bookings. The sign shows the next seven days, today first, each with its name, its date and its bookings (the time, the title and Zoom or Google Meet); a day with nothing booked says **Nothing booked**. The seven days run across the screen, or one under another on a sign standing upright, where a booking's title is cut after one line. On a screen lying down and 700 pixels high or less a booking is its time and one line of title, and on one narrower than 1000 pixels the days are named by three letters (**Wed**). A booking is listed once, on the day it starts, and an all-day booking shows 12:00 AM. The sign never scrolls: where a day has more bookings than fit, it shows the first ones and **+3 more**. The week is read from the device when you open it, so open it again to see a change. **Back**, or thirty seconds without a tap, returns to Today. If the week cannot be read the sign says **The week could not be loaded.** and **Back** still works.
- **Football scores, off by default.** On the table screen tap **Options**, then **Door sign**, and under **Football scores** switch on any of **Texas**, **Michigan**, **Texas A&M** and **Missouri** (the ones that are on are blue). A strip appears along the bottom of the door sign. On its left is the school's last result, with the school's own score first whether it played at home or away (**Texas (#4) 45 – Texas A&M 42**). On its right is the next game, the visiting team first (**Next: 10/19 · Texas (#4) @ Missouri (#9)**; a game at a neutral site reads **vs** in place of **@**). A number in orange is a team's place in the Top 25. With more than one school on, the strip shows one at a time and moves to the next every 8 seconds, with the school's name at the start of the line (**Michigan: Michigan (#12) 31 – Ohio State 24**). Switch every school off and the strip goes. Nothing else on the sign moves when the strip comes or goes, except on a short screen (below). The strip never wraps: a name too long for its line ends in "…", and on a screen narrower than 1100 pixels the last result sits above the next game.
- **Where the scores come from.** The device, not the door screen, asks ESPN's public schedule feed at `site.api.espn.com` for each chosen school every ten minutes, and asks for nothing while every school is off. There is nothing to set up: no account and no key. The device needs to reach `site.api.espn.com` over the internet on HTTPS (port 443); nothing else on the door sign needs the internet. The request names only the school and says nothing about the room or the people in it. ESPN does not document this feed or promise to keep it: it can change or stop, and the strip then keeps showing the last result it read until Crystal PM updates the software.
- **On a short screen.** The label under the room's name, the line under it and the room's location are cut with an ellipsis after two lines, and so is a booking's title (after one line in Today on a screen 560 pixels high or less). A room name that would take more than two lines is set in smaller type, as one of more than 24 characters is, so it is never cut; on a screen 560 pixels high or less, one that would take two lines is too, and one that still takes more than two goes on down 0.1 rem at a time, to 1.3 rem, until it takes two. On a screen of any height a name also goes down, to 1.3 rem at the most, while the status would run off the bottom. At 700 pixels high or less the week leaves out the line under the status label and the room's location and makes the room's name smaller, and at 560 pixels or less each day's date too; with football scores on, at 560 pixels high or less, Today's status leaves out the line under its label too.

Book the room

The room books itself in its Google calendar from the door sign and from the table screen, for in-person meetings that need no Zoom or Meet link. It needs the room's calendar shared with the service

account with **Make changes to events** (the guide "Connect Crystal Meet rooms to Google Calendar"); `croom --check-calendar -c /etc/croom/config.yaml` says whether the room can book.

- **Book.** Tap **Book** under the status on the door sign, or **Book room** on the table screen. Choose the day (today and the six days after it), then one of its open start times, every 15 minutes from 8:00 AM to the last start before 6:00 PM. Type your name or the meeting's name, choose how long from the list (15 minutes to 4 hours, in steps of 15, as long as the time allows before the next booking and 6:00 PM), and tap **Book**. The booking is in Google Calendar at once, with the room as its calendar and "Booked on the <room> door sign" in its description. Another booking made in Google since the screen last looked is refused with its name.
- **Use now.** Walk in to a free room and tap **Use now**: type your name, choose how long, and tap **Start**. The room is booked from this minute, the door sign and the panel's light bar turn red, and the booking is in Google Calendar. Use now is offered while the room is free for at least 15 minutes, and is not held to the booking hours.
- **End now and Cancel.** A meeting made at the room's screens that is in progress has **End now** on the table screen (never on the door sign, so nobody in the hallway ends a meeting): tap it, then **Tap again to end**, and the booking is shortened to this minute in Google Calendar. One that has not started has **Cancel** on its row in Today, on the door sign and on the table screen, also with a second tap. Meetings booked in Google Calendar by people are only ever changed there.
- **When Google cannot be reached.** The room keeps working: what is booked, started, ended or cancelled at its screens is kept on the device (`bookings.json` in its data folder) and sent to Google Calendar on the first poll that gets through, in order. The screens show the day as Google last gave it (`calendar-cache.json` , kept for a restart too) with the room's own bookings over it, and say **Calendar offline since 10:42 AM. Bookings made here will sync.** A booking of the room's that has not reached Google yet says **Not in Google Calendar yet**.
- **Double bookings.** While Google is out of reach, somebody may book the same time in Google Calendar. When it is back, the room's booking is written all the same, and shows **Double-booked with** the other booking's name on both screens until somebody moves or cancels one of them. `croom --check-calendar` lists them, and what is still waiting to be sent.
- **Other hours.** To offer other start times, set `booking_start` and `booking_end` under `calendar` in `/etc/croom/config.yaml` , as `"07:30"` and `"19:00"` for example (on the quarter hour), and restart the service.
- **Leaving the steps.** **Back** returns a step; left alone for a minute the steps close by themselves. On the table screen the room page's own keyboard comes up for the name, and its **Done** books; the door panel brings up Android's.

Photograph credits

The eight photographs are from Wikimedia Commons. They are resized and shown under the navy overlay, each under its author's licence; `CREDITS.md` in the repository has the file names and links.

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IF SOMETHING IS OFF

Troubleshooting

The room page does not load

On the Pi, type `sudo systemctl status croom`. If it is not running, type `sudo journalctl -u croom -n 50` and look for the first line marked ERROR. Make sure port 8080 is not used by something else on the device.

The device shows offline

From the Pi, type `curl http://<dashboard address>:3001/health`. If nothing comes back, the dashboard is not reachable from the room's network: check the address in `~/room.yaml` and that the dashboard Pi is up (its guide has the checks). If the token was already used, create a new one and run the installer again with the updated config.

The browser opens but never joins

Zoom needs the meeting's passcode: paste the full invitation link, or type the passcode in the **Passcode** field that appears for a meeting ID or a link without one (on a "Couldn't join" card that says "Zoom needs the meeting passcode", press **Join again**). Google Meet needs the room signed in (the guide "Let Crystal Meet rooms join Google Meet"); the room page says so when the sign-in is missing or expired. The room page quotes what Meet showed and the log names a screenshot under `/tmp`; from the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` reports the same (stop the service first: `sudo systemctl stop croom`).

No sound or picture in the call

Check the USB camera and speakerphone are connected before the device starts, then reboot the Pi. The service runs as the signed-in desktop user so it can use the room's screen and audio. From the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` (stop the service first: `sudo systemctl stop croom`) prints "Devices the browser sees" with the microphones, speakers and cameras the browser can use; if the MeetUp is missing from "Devices the browser sees" but `wpctl status` lists it, the browser cannot reach PipeWire: make sure `libpulse0` is installed (the installer adds it) and that the service runs in the desktop user's session.

No browser window opens on the TV

The service waits for the desktop before it starts, so give a freshly booted Pi a minute. If the screensaver still does not appear, type `sudo journalctl -u croom -n 50` and look for "Failed to initialize". The browser is installed under `/opt/croom/browsers`; if that folder is missing, run the installer again.

The screens cannot open the page

Use the device's IP address rather than its `.local` name; many tablets do not resolve those. Check that the screen and the device are on the same network, then open `http://<device address>:8080/api/status` from a laptop to confirm the device answers.

The TV is not found

Find TVs lists nothing, or the table says "The TV is not answering at 10.0.0.41". Each TV answers the device's search straight back to the device, so a firewall on the device that blocks incoming traffic by default hides every TV: allow incoming UDP from the TV's address, or skip Find TVs and read the address in the TV's network settings. The search and the wake-up messages leave by the device's default network connection, so the device and the TV must be on that same network. Check that the TV is on, that **LG Connect Apps** is on and that **host** is the TV's current address, then type `/opt/croom/venv/bin/croom --check-tv -c /etc/croom/config.yaml` to see what the device sees.

The TV will not wake

Turn on says the TV did not turn on, or a join goes ahead with the TV dark. A MAC address in **mac** that is missing or is not one stops every wake, Quick Start+ or not, and the check warns about it. A TV that is fully off, with Quick Start+ off, can only be woken by Wake-on-LAN: **Mobile TV On** must be on and the MAC address must be the TV's own (a TV that has a cable and Wi-Fi has one for each). With **Quick Start+** on, a TV that answers is told to turn on over its connection, so Mobile TV On and a wrong but well-formed MAC address do not matter. A join waits up to 15 seconds for the TV, with the page showing Joining, and then goes ahead without it.

The TV no longer accepts the table

The table says "The TV no longer accepts this table" or, on the remote's top line, "The TV is asking to pair again", or the TV keeps showing a request to allow a device: it has forgotten this table, after a reset or when its list of paired devices was cleared. Tap **Forget TV** (on the remote it asks for a second tap), then **Pair**, and accept the new request on the TV with its own remote. While the TV shows a request the device waits for an answer; it stops waiting when the request is answered, when the TV goes away, or on Forget TV.

The room Pi's mouse does nothing

Pointer moves the TV's cursor but **Room Pi mouse** moves nothing, and the table says the mouse "is not allowed to use /dev/uinput" or "needs /dev/uinput, and it is not there". The device makes its virtual mouse on `/dev/uinput`, which needs `python3-dev`, `gcc` and `libc6-dev` on the device before the software is installed (pip builds the mouse library), this udev rule, `KERNEL=="uinput", GROUP="input", MODE="0660"`, in `/etc/udev/rules.d/99-croom-uinput.rules`, the service's user in the `input` group, and the `uinput` module loaded; the installer's virtual-mouse step sets up the last three. Check with `ls -l /dev/uinput` (group input), `groups pi` (your user name instead of pi) and `lsmod | grep uinput`. Run the virtual-mouse step from the update recipe above, then `sudo systemctl restart croom`. The mouse is made when the first touchpad mode is turned on after the service starts; its pointer should then appear on the TV.

The football strip does not appear

A school you switch on shows within a few seconds. If nothing comes, the device is not getting an answer from ESPN. From the Pi, `curl -s -o /dev/null -w '%{http_code}\n' https://site.api.espn.com/apis/site/v2/sports/football/college-football/teams/251/schedule` prints 200 when it can; if it does not, look at the office network for a firewall or web filter that blocks `site.api.espn.com` on HTTPS. `sudo journalctl -u croom -n 50` names the school and the reason (look for "football schedule"). It says so when a school starts to fail, not at every try; the device tries again after 1, 2 and 5 minutes and then every 10, and the strip comes back by itself when a read works. While ESPN cannot be reached the strip keeps the last result it read, so a game played since can still show as next. The strip shows the last game ESPN lists as final and the next one it lists as scheduled, so between seasons it may show the last result alone, for months, until ESPN lists the new schedule.

A choice is lost when the device restarts

The table screen's choices (the background, the football switches and the screensaver) are saved in `control-settings.json` in the device's data folder, `/var/lib/croom` (a `croom` folder under the desktop user's `.local/share` when the service cannot write there). If the file cannot be written, because the SD card is full or read-only or the folder is not the desktop user's, the choices still work, but they are lost at the next restart until the folder can be written. `sudo journalctl -u croom -n 50` says "Could not save the room settings" and why; `df -h /var/lib/croom` shows whether the card is full and `ls -ld /var/lib/croom` who owns the folder (the installer gives it to the desktop user).

A name too long even at the smallest size

A room name goes down to 1.3 rem and no further. Room names over about 150 characters (five lines of the smallest type on a 7-inch 800 by 480 sign) run off the sign; use a shorter name.

GOOD TO KNOW

Under the hood the device software is still called `croom` : that is the name you will see in service and log commands. Everything people see says Crystal Meet.